Shijian Xu

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Education

- **Universität Basel** — **Basel, Switzerland**
  Ph.D. student — 04/2024 – 04/2028 (expected)
  Supervisor: Prof. Volker Roth
- **EPFL** — **Lausanne, Switzerland**
  M.Sc. in Data Science — 09/2020 – 02/2023
  Thesis: Weight Averaging for Out-of-Distribution Generalization and Few-Shot Domain Adaptation
  Supervisor: Prof. Grigorios Chrysos & Prof. Volkan Cevher
- **Nanjing University** — **Nanjing, China**
  B.Sc. in Computer Science and Technology — 09/2014 – 06/2018
  Thesis: Predicting Smartphone App Usage with Recurrent Neural Networks
  Supervisor: Prof. Wenzhong Li

Publications & Preprints

- Temporal Self-Distillation: Faster Inference in Discrete Diffusion Language Models
  Shijian Xu*, Andrea Miele*, Metod Jazbec, Volker Roth, Eric Nalisnick, Ilija Bogunovic
  NeurIPS 2026 Workshop DiffuLM, *`equal contribution`
- NIMO: a Nonlinear Interpretable MOdel
  Shijian Xu, Marcello Massimo Negri, Volker Roth
  ICLR 2026
- Simple Control Baselines for Evaluating Transfer Learning
  Andrei Atanov*, **Shijian Xu***, Onur Beker, Andrei Filatov, Amir Zamir
  arXiv, 2022
- Predicting Smartphone App Usage with Recurrent Neural Networks
  S. Xu, W. Li, X. Zhang, S. Gao, T. Zhan, Y. Zhao, W.-W. Zhu, T. Sun
  International Conference on Wireless Algorithms, Systems, and Applications (WASA), 2018

Experience

- **University of Basel** — **Basel, Switzerland**
  Ph.D. research — 08/2024 – Current
  - Interpretable ML: a nonlinear yet interpretable model that preserves the transparency of linear models while capturing the complex representations enabled by neural networks, paper accepted in ICLR 2026
  - Accelerating diffusion LLMs (paper in submission)
- **LIONS Lab, EPFL** — **Lausanne, Switzerland**
  Master Thesis, supervisor: Dr. Grigorios Chrysos & Prof. Volkan Cevher — 09/2022 – 02/2023
  Out-of-Distribution Generalization
  - Improved weight averaging based method for OOD generalization and few-shot domain adaptation
- **DisneyResearch|Studios** — **Zürich, Switzerland**
  Research Intern, mentor: Dr. Christopher Schroers & Dr. Abdelaziz Djelouah — 02/2022 – 08/2022
  - Image super-resolution: develop and train a deep learning-based model for internal use
- **VILAB, EPFL** — **Lausanne, Switzerland**
  Semester Project, supervisor: Prof. Amir Zamir — 03/2021 – 12/2021
  Evaluation of Self-Supervised Learning
  - Proposed an evaluation standard for measuring the effectiveness of pre-training methods

- **Tencent PCG** — **Shanghai, China**
  Research Intern — 07/2021 – 09/2021
  - Transformer-based multi-label image classification
- **Tencent AI Lab** — **Shenzhen, China**
  Research Intern — 02/2020 – 08/2020
  - Self-supervised multi-modal video classification
- **City University of Hong Kong** — **Hong Kong**
  Researcher — 09/2018 – 12/2019
  - Deep learning for image specular highlight removal
  - Semi-supervised single image heavy rain removal
- **Nanjing University** — **Nanjing, China**
  Project Student, supervisor: Prof. Wenzhong Li — 09/2015 – 06/2018
  - LSTM based model for app usage prediction and recommendation

Teaching and Service

Teaching assistant on the following courses:

- Scientific Computing (University of Basel) — 2024, 2025, 2026
- Seminar: Machine Intelligence (University of Basel) — 2025, 2026

Awards

- Second Prize of Excellent Bachelor Thesis Award, Nanjing University — 2018
- Third Prize of Elite Program Scholarship, Nanjing University — 2017
- Third Prize of People's Scholarship, Nanjing University — 2016
- Second Prize of People's Scholarship, Nanjing University — 2015

Languages

- **English:** Advanced, TOEFL iBT 103 (2023)
- **German:** Beginner, B1
- **French:** Beginner, A1/A2
- **Chinese:** Native